Quantum Mechanics I — Exercise Sheet 10

Two systems

Variant: Questions only

Conventions, as in Lecture 10: a vector of the pair is $\psi = \sum_{i,j} C_{ij}\, e_i \otimes e_j$, with the coefficient matrix C and the first factor the slow index; $|jk\rangle = |j\rangle \otimes |k\rangle$, $|0\rangle = |\uparrow\rangle$; $\psi^- = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ and $t_0 = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$; the swap $W(u \otimes w) = w \otimes u$.

Exercise 1 (product or entangled). Decide whether each state is a product, and if it is, find the factors.

(a) $\frac{1}{2}\big(|00\rangle + \mathrm{i}\,|01\rangle - |10\rangle - \mathrm{i}\,|11\rangle\big)$ on two qubits.

(b) $\frac{1}{\sqrt{3}}\big(|00\rangle + |01\rangle + |10\rangle\big)$ on two qubits.

(c) $\frac{1}{\sqrt{10}}\big(e_0 \otimes (e_0 + e_1) + e_1 \otimes (2e_0 + 2e_1)\big)$ in $\mathbb{C}^2 \otimes \mathbb{C}^3$, with the bases e_0, e_1 and e_0, e_1, e_2.

(d) $\frac{1}{\sqrt{2}}\big(e_0 \otimes e_0 + e_1 \otimes e_2\big)$ in $\mathbb{C}^2 \otimes \mathbb{C}^3$.

Exercise 2 (how entangled two qubits can be). Let ψ be a unit vector of two qubits with the coefficient matrix C. Show that $|\det C| \le \frac{1}{2}$. Check that ψ^-, the state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, and the state of hydrogen's two spins in Example 10.6 at $\nu t = \pi/2$ reach the bound.

Exercise 3 (a reading of the pair entangles).

(a) Write the 4×4 matrix of $\boldsymbol{\sigma} \cdot \boldsymbol{\sigma} = \sum_a \sigma_a \otimes \sigma_a$ in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, and check that it is $2W - \mathbb{1}$.

(b) The swap W is an observable of the pair, with the readings ± 1. Two spins are prepared in the product $|01\rangle$ and W is read. Find the probabilities of the readings and the states afterwards.

Exercise 4 (two precessions). Two spins do not interact. The first precesses about z and the second about x: $H_1 = \frac{\hbar\nu_1}{2}\sigma_z$ and $H_2 = \frac{\hbar\nu_2}{2}\sigma_x$, with $\nu_1, \nu_2 > 0$. The pair starts in $|00\rangle$. Find the state at time t, and the probability that both spins read up along z. Is the state ever entangled?

Exercise 5 (hydrogen in a weak field). Hydrogen's two spins in a field along z have the Hamiltonian (10.4). Treat the field as a perturbation: $H_0 = \frac{\hbar\nu}{4}\boldsymbol{\sigma} \cdot \boldsymbol{\sigma}$ and $H_1 = \frac{\hbar\nu_e}{2}\sigma_z \otimes \mathbb{1} - \frac{\hbar\nu_p}{2}\mathbb{1} \otimes \sigma_z$.

(a) Find $\|H_1\|$ and the gap of the triplet level $\hbar\nu/4$ of H_0. For which fields does Theorem 6.8 apply to that level? Give the largest B, with $\nu_e/2\pi = 28.0$ GHz and $\nu_p/2\pi = 42.6$ MHz per tesla.

(b) Find the eigenvalues and eigenvectors of EH_1E on the triplet, E its projection. What does the theorem say about the three levels near $\hbar\nu/4$?

(c) Compare with Proposition 10.7. Which first-order energies are exact? By how much does the $m = 0$ level of the triplet move, and is this within the theorem's bound?

Exercise 6 (⋆ correlations of products). Consider three ensembles of pairs of qubits: the z-ensemble $\{(\frac{1}{2}, |01\rangle), (\frac{1}{2}, |10\rangle)\}$ of § 10.4, the x-ensemble $\{(\frac{1}{2}, |{+}x\rangle \otimes |{-}x\rangle), (\frac{1}{2}, |{-}x\rangle \otimes |{+}x\rangle)\}$, and the ensemble that takes a member of either with probability $\frac{1}{2}$. For each, find $\langle \sigma_z \otimes \sigma_z \rangle$ and $\langle \sigma_x \otimes \sigma_x \rangle$, the means $\mathrm{Tr}\big(\rho(\sigma_a \otimes \sigma_a)\big)$. Compare with the bound of § 10.4 and with ψ^-.